# ICT Course: Introduction to Cryptography

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# Session 7: Digital signature and Hash function

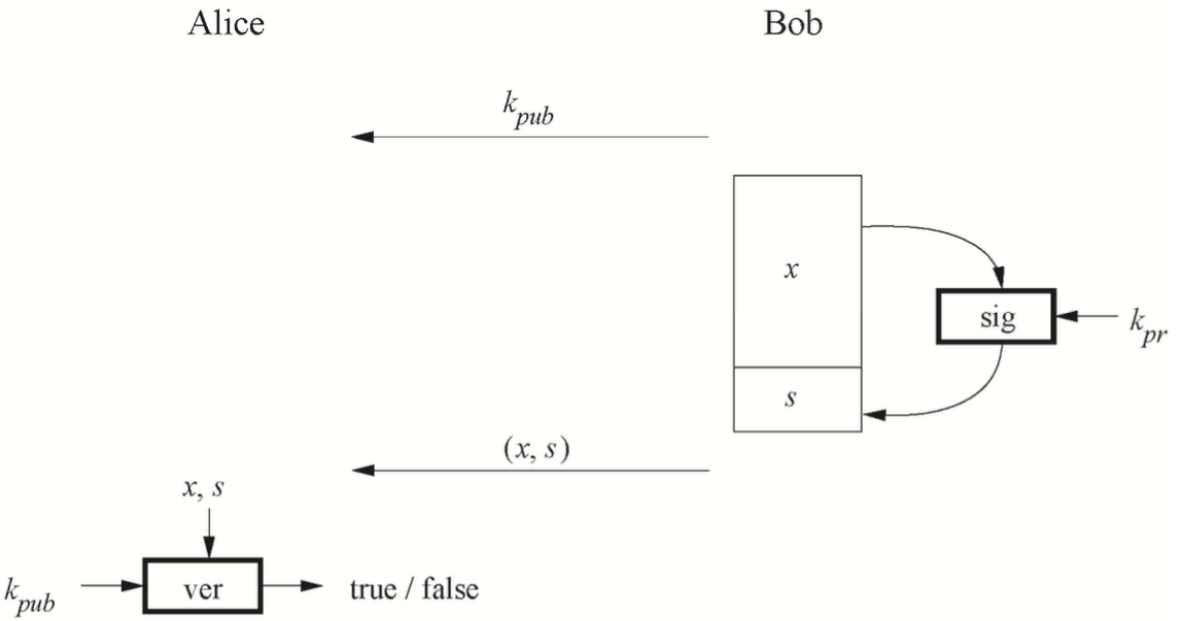

1. Digital Signature
   - Principles of Digital Signatures
   - RSA signature scheme
   - Elgamal Digital Signature
   - Digital signature algorithm(DSA)
2. Hash function
   - Overview
   - Security requirements of Hash functions
   - Hash function algorithms

# Digital Signature

Digital signature provides:

- Message authentication: assures that message is authentic to one user (the same functionality of hand-written signature)
- Integrity
- Nonrepudiation

# Principles of Digital Signatures




- Key generation?
- Signature Algorithm?
- Verification function?

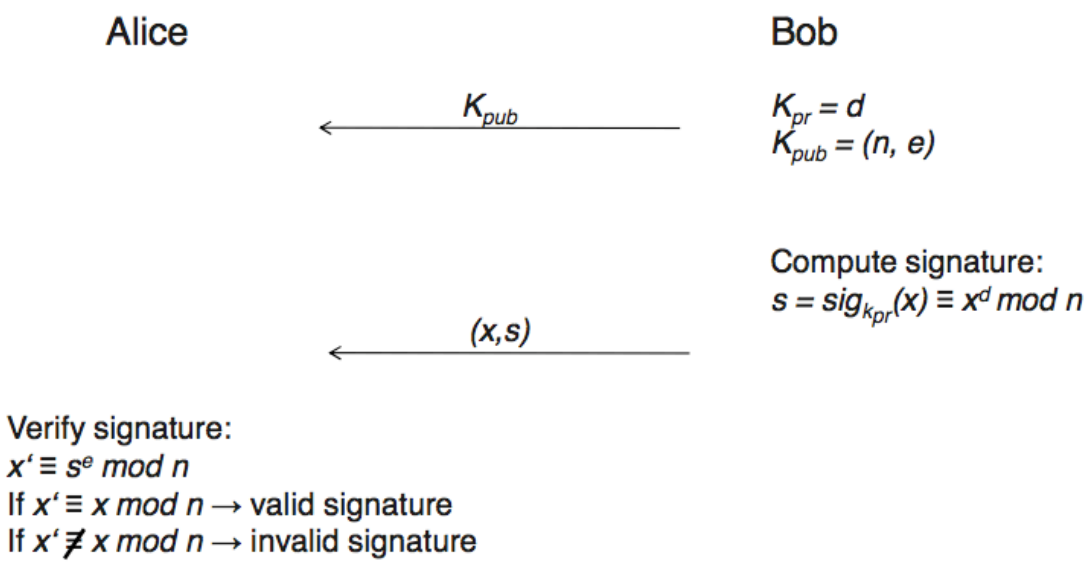
# RSA signature scheme

- Generate private keys and public keys: use the RSA key generation
- Generate signature: "encrypt" the message with private key to obtain $s$, append $s$ to the message
- Verify signature: "decrypt" the signature with public key and compare it with the message

# RSA signature scheme - Details

Key generation:

- Private key: $k_{pr} = d$
- Public key: $k_{pub} = (n, e)$




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Chapter 10 of *Understanding Cryptography* by Christof Paar and Jan Pelzl

# RSA Signature Scheme - Security

- $n$ needs to be at least 1024 bits
- Forgery Attack:
  - Trudy can generate a valid signature for a random message
  - Method:
    1. Choose a signature
    2. Compute the respectively message from $k_{pub}$
  - Trudy can claim to Alice that he's Bob
- Solution against Forgery Attack: Padding Scheme

# Elgamal Digital Signature

- Key generation?
- Signature generation and Verification
- Example: Bob wants to send a message to Alice. This time, it should be signed with the Elgamal digital signature scheme. Describe the key generation, signature generation and verification process.

$$p = 29, \alpha = 2, d = 12, k_E = 5$$

# Digital Signature Algorithm (DSA)-Overview

- Federal US Government standard for Digital Signature
- Proposed by NIST
- Based on Elgamal signature scheme
- Signature length: 320 bits
- Signature verification is slower compared to RSA

# DSA - Key Generation

Key generation:

1. Generate a prime $p$: $2^{1023} < p < 2^{1024}$
2. Find a prime divisor $q$ of $p-1$: $2^{159} < q < 2^{160}$
3. Find $\alpha$: $ord(\alpha) = q$
4. Choose a random integer $d$: $0 < d < q$
5. Compute $\beta \equiv \alpha^d \ mod \ p$

Key pairs:

- $k_{pub} = (p, q, \alpha, \beta)$
- $k_{pr} = d$

# DSA - Signature Generation - Verification

Signature Generation

1. Choose random ephemeral key $k_E$: $0 < k_E < q$
2. Compute $r \equiv (\alpha^{k_E} \ mod p) \ mod \ q$
3. Compute $s \equiv (SHA(x) + d * r) * k_E^{-1} \ mod \ q$

Signature Verification

1. Compute $w \equiv s^{-1} \ mod \ q$
2. Compute $u_1 \equiv w * SHA(x) \ mod \ q$
3. Compute $u_2 \equiv w * r \ mod \ q$
4. Compute $v \equiv (\alpha^{u_1} * \beta^{u_2} \ mod \ p) \ mod \ q$
5. Verification $ver(x, (r, s))$:
   if $v \equiv r \ mod \ q$, then valid signature
   if $v \not\equiv r \ mod \ q$, then invalid signature

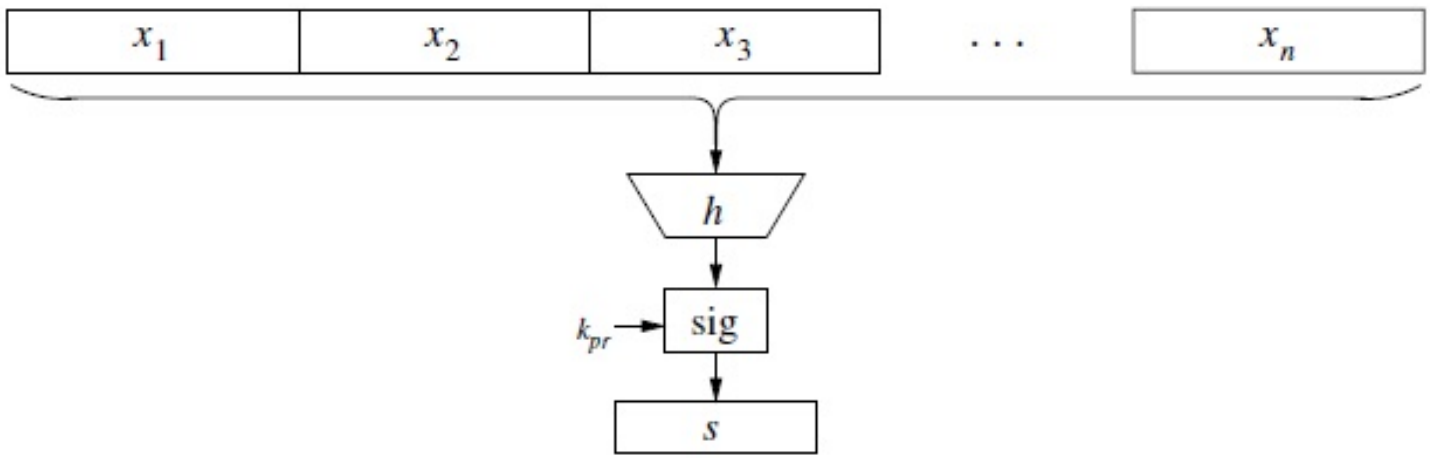
# DSA - Example

Bob wants to send a message x to Alice which is to be signed with the DSA algorithm. Suppose the hash value of x is
$h(x) = 26, p = 59, q = 29, \alpha = 3, d = 7, k_E = 10$. What is the key pair? Describe the signature generation and verification.

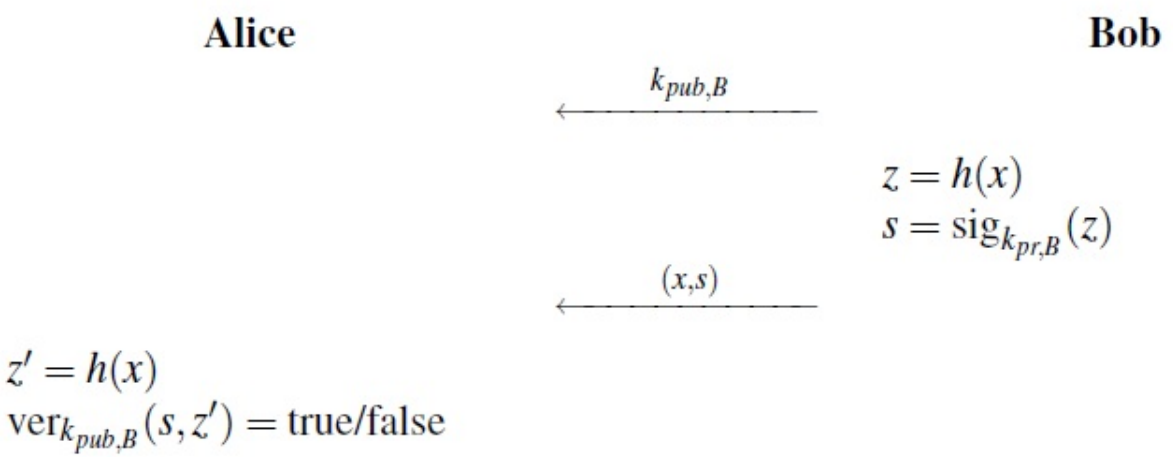
# Hash function - Why it is required?

Signing a long message:

- If the message is divided into block size that is less than allowed input size of digital signature algorithm
  - High computational load
  - Message overhead
  - Security limitations
- a short signature for the message: hash function



# Basic protocol for Digital Signature with a hash function




Desirable properties of hash function:

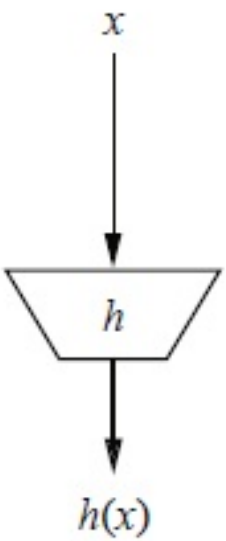
- Fast to compute
- Output is fixed length and independent of input length
- Computed fingerprint (output of the hash function) is highly sensitive to all input bits (a minor modification of input leads to different output)

# Security requirements of hash functions

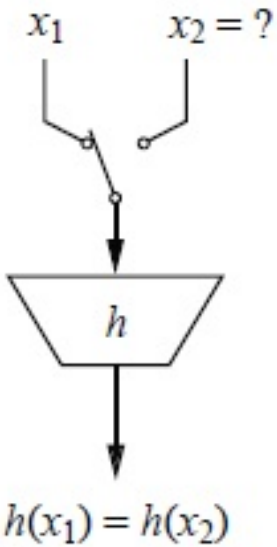
Properties needed for a hash function to be "secure":

1. preimage resistance (one-wayness): h(x) is one-way
2. second preimage resistance (weak collision resistance):
   $\nexists x_2 \neq x_1 : h(x_1) = h(x_2)$
3. collision resistance (strong collision resistance):
   $\nexists h(x_1) = h(x_2) : x_1 \neq x_2$

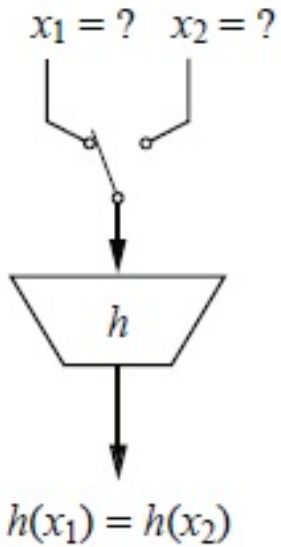
# Security requirements of hash functions




preimage resistance




second preimage resistance




collision resistance

# Hash function algortihms

Two general types of hash functions:

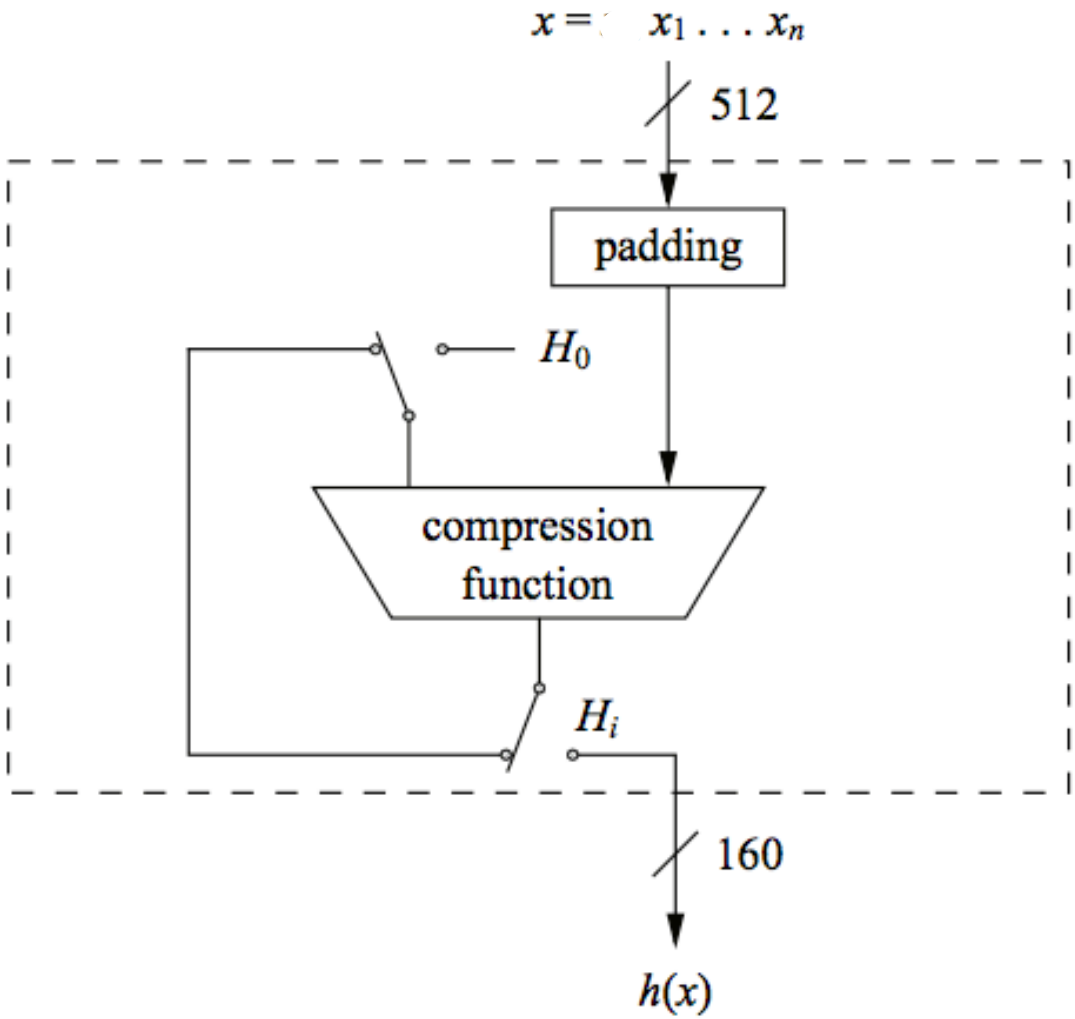
- Dedicated hash functions: algorithms are specifically designed to serve as hash functions, e.g., MD4 family: MD5, SHA-1...
- Block cipher hash functions: using block cipher to construct hash functions

# Hash Algorithm SHA-1

- part of MD4 family
- based on Merkel-Damgard construction
- 160-bit output from a message of maximum length $2^{64}$ bits
- Widely used

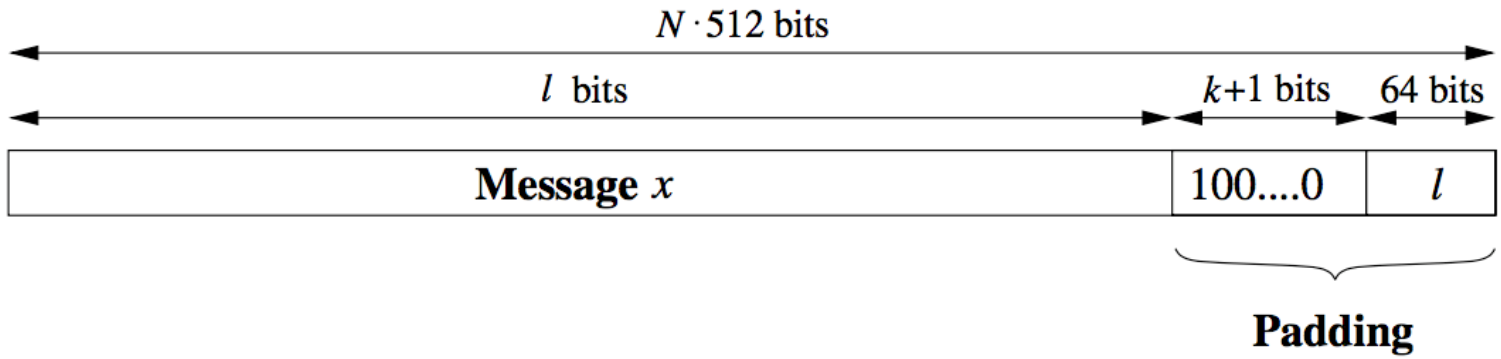
# SHA-1 High-level Diagram

Compress function: 80 rounds, 20 rounds/stage



# SHA-1 Prepocessing

- Padding:
  - Message length: $l$ bits
  - Padded message length: multiple of 512 bits




- Dividing the padded message: 512 bit blocks $x_i$, each block 16 $words$ $x$ 32 $bits$
- Initial fixed hash value $H_0$: 160 bits $\equiv$ 5 words $H_0^{(i)}$

# SHA-1 Hash computation

- Message schedule: 80 words $W_j$ are derived from 16 words of a block in the padded message

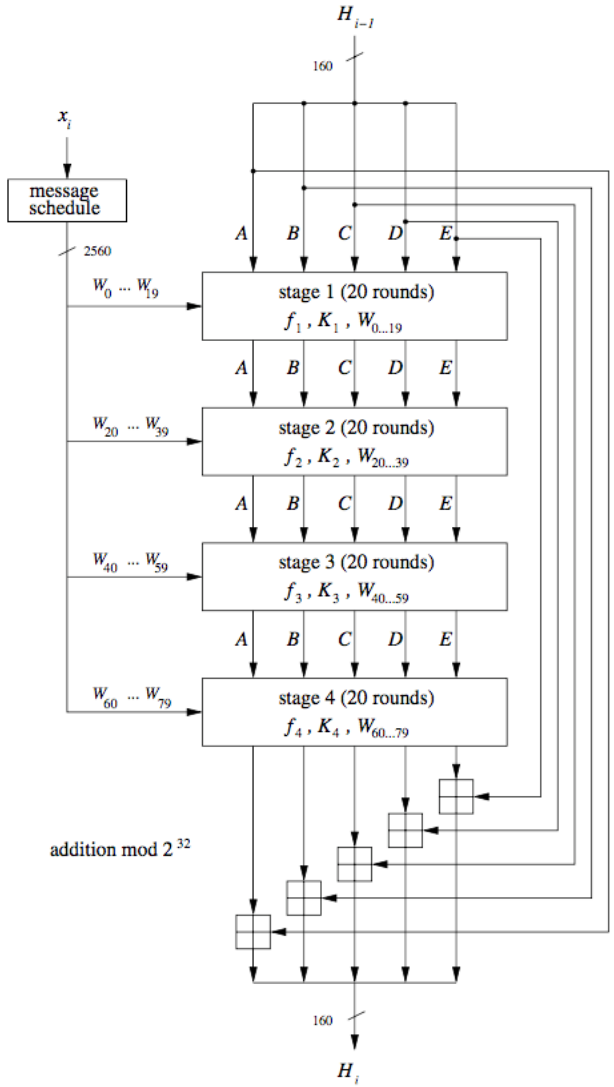
$$W_j = \begin{cases} x_i^{(j)} & \text{if } 0 \le j \le 15 \\ (W_{j-16} \oplus W_{j-14} \oplus W_{j-8} \oplus W_{j-3})_{<<<1} & \text{if } 16 \le j \le 79 \end{cases}$$

denote: $X_{<<<n}$: circular left shift of the word $X$ by $n$ bit positions.

# SHA-1 Hash computation - Compression function

80-round compression function:

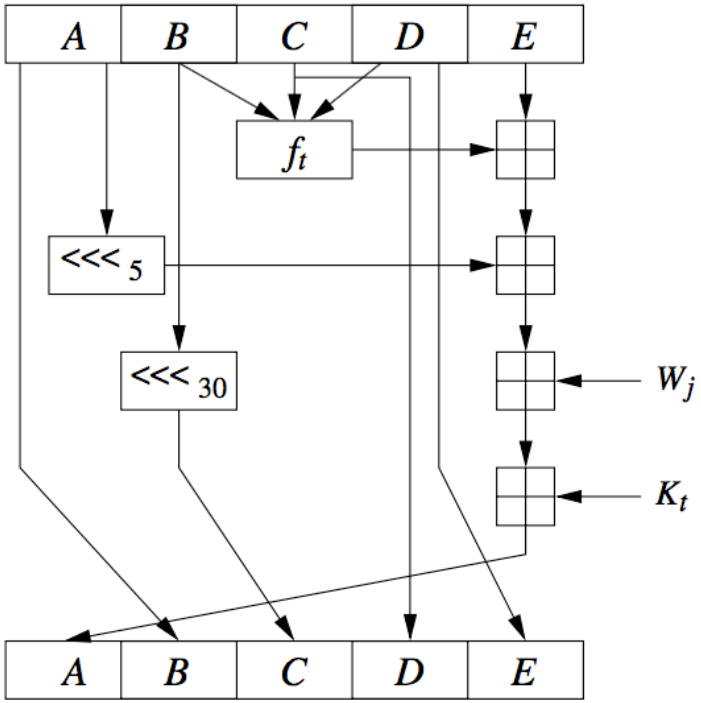
- In each stage $t$ (20 rounds): internal function $f_t$, constant $K_t$ are specified
- output after 80 rounds is added to the input value $H_{i-1}$ modulo $2^{32}$ in word-wise fashion.



# SHA-1 Hash function - Compression function

The operation within round $j$ in stage $t$:

$$A, B, C, D, E = (E + f_t(B, C, D) + (A)_{<<<5} + W_j + K_t), A, (B)_{<<<30}, C, D$$



# SHA-1 - Parameters

Initial hash values:

$$A = H^{(0)}{}_0 = 674523010$$

$$B = H^{(1)}{}_0 = EFCDAB890$$

$$C = H^{(2)}{}_0 = 98BADCFE0$$

$$D = H^{(3)}{}_0 = 103254760$$

$$E = H^{(4)}{}_0 = C3D2E1F0.$$

Round functions and round constants for the SHA rounds

| Stage $t$ | Round $j$ | Constant $K_t$ | Function $f_t$ |
|---|---|---|---|
| 1 | 0...19 | $K_1 = \texttt{5A827999}$ | $f_1(B,C,D) = (B \wedge C) \vee (\bar{B} \wedge D)$ |
| 2 | 20...39 | $K_2 = \texttt{6ED9EBA1}$ | $f_2(B,C,D) = B \oplus C \oplus D$ |
| 3 | 40...59 | $K_3 = \texttt{8F1BBCDC}$ | $f_3(B,C,D) = (B \wedge C) \vee (B \wedge D) \vee (C \wedge D)$ |
| 4 | 60...79 | $K_4 = \texttt{CA62C1D6}$ | $f_4(B,C,D) = B \oplus C \oplus D$ |

# Exercise

Compute the output of the first round of stage 1 of SHA-1 for a 512-bit input block of

1. $x = 0...00$
2. $x = 0...01$ (i.e., bit 512 is one).

Ignore the initial hash value $H_0$ for this problem (i.e., $A_0 = B_0 = ... = 00000000hex$).